TASK 4 – SEVERITY & CASE CLASSIFICATION

VirtualWorks Lab | Pharmacovigilance Internship | Task ID: task4

1. Objective

To classify adverse drug reaction (ADR) cases according to seriousness and expectedness, and explain the clinical and regulatory basis for each classification.

2. Key concepts

Seriousness is a regulatory classification based on the outcome of an event. Serious criteria include death, a life-threatening event, inpatient hospitalization or prolongation of hospitalization, persistent or significant disability, congenital anomaly/birth defect, or another medically important event.

Severity describes the intensity of a reaction (mild, moderate or severe). Severity and seriousness are not interchangeable.

Expectedness describes whether the nature, specificity and severity of the reaction are consistent with the applicable reference safety information for the particular medicinal product, such as its approved label or SmPC.

3. Case classification summary

Case	Suspected medicine	Reaction	Seriousness	Expectedness
01	Amoxicillin	Skin rash and itching	**Non-serious***	**Expected***
02	Ibuprofen	Gastrointestinal bleeding requiring hospitalization	**Serious**	**Expected***
03	Metformin	Lactic acidosis	**Serious**	**Expected***

*Expectedness is a preliminary educational classification based on the general known safety profile. Formal expectedness requires comparison with the applicable product-specific reference safety information. Case 1 is considered non-serious only because no serious outcome or medically important complication is described.

4. Individual case assessments

Case 01 – Amoxicillin: Skin rash and itching

Seriousness: Non-serious (provisional). The case reports skin rash and itching but does not mention death, life-threatening consequences, hospitalization, disability, congenital anomaly or another medically important event. If the rash represented a severe cutaneous adverse reaction or required hospitalization, the assessment would need revision.

Expectedness: Expected (provisional). Skin rash is a recognized adverse reaction associated with amoxicillin. The final determination must be checked against the specific product's approved prescribing information or SmPC.

Case 02 – Ibuprofen: Gastrointestinal bleeding requiring hospitalization

Seriousness: Serious. The case explicitly states that gastrointestinal bleeding required hospitalization. Inpatient hospitalization is a regulatory seriousness criterion.

Expectedness: Expected (provisional). Gastrointestinal ulceration and bleeding are recognized risks described in ibuprofen product information. The exact reference safety information for the product involved should be checked.

Case 03 – Metformin: Lactic acidosis

Seriousness: Serious. Lactic acidosis is potentially life-threatening and is a medically important event. Metformin product information includes a warning about lactic acidosis.

Expectedness: Expected (provisional). Lactic acidosis is a recognized risk described in metformin prescribing information. The specific product label or SmPC should be consulted for formal expectedness assessment.

5. Overall interpretation

The three cases demonstrate that seriousness and expectedness are separate assessments. A reaction can be serious and expected at the same time. For example, ibuprofen-associated gastrointestinal bleeding requiring hospitalization is serious because hospitalization occurred, while gastrointestinal bleeding is also a recognized risk of ibuprofen.

Based on the descriptions visible in the Task 4 assignment, the preliminary classifications are: Case 1 – non-serious and expected; Case 2 – serious and expected; Case 3 – serious and expected. These classifications should be updated if additional clinical details or product-specific reference safety information are supplied.

6. Conclusion

The assessment of seriousness considers the regulatory outcome or medical importance of the reaction, whereas expectedness is determined by comparison with the applicable product-specific safety information. Correctly distinguishing these concepts supports consistent pharmacovigilance documentation, case evaluation and reporting decisions.

7. References

1. International Council for Harmonisation (ICH). *E2A: Clinical Safety Data Management – Definitions and Standards for Expedited Reporting.* ICH E2A guideline

2. U.S. Food and Drug Administration. Amoxicillin prescribing information. FDA label PDF

3. U.S. National Library of Medicine. DailyMed – Ibuprofen prescribing information. DailyMed

4. U.S. National Library of Medicine. DailyMed – Metformin prescribing information. Metformin label